

Math 521
Homework 3

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Problem 1. Modify the proof of the mean-value formulas to show for $n \geq 3$ that:

$$u(0) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(0,r)} g \, dS + \frac{1}{n(n-2)\alpha(n)} \int_{B(0,r)} \left(\frac{1}{|x|^{n-2}} - \frac{1}{r^{n-2}} \right) f \, dx$$

provided that:

$$\begin{cases} -\Delta u = f, & \text{in } B^o(0, r), \\ u = g, & \text{on } \partial B(0, r). \end{cases}$$

Proof. Define:

$$\begin{aligned} \rho(r) &:= \int_{\partial B(0,r)} u \, dS \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(0,r)} u \, dS \\ &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} u(r\omega) \, d\omega. \end{aligned}$$

Using the same technique from the proof of the mean-value formula gives:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} \frac{d}{dr} u(r\omega) \, d\omega \\ &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} Du(r\omega) \cdot \omega \, d\omega. \end{aligned}$$

Making the substitution $y = r\omega$ yields $dy = r^{n-1} d\omega$ and $\omega = \frac{y}{r}$. Hence:

$$\rho'(r) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(0,r)} Du(y) \cdot \frac{y}{r} \, dy.$$

Note that $\frac{y}{r}$ is the outward normal on S^{n-1} . Using Green's Formula I gives:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)r^{n-1}} \int_{B(0,r)} \Delta u(y) \, dy \\ &= -\frac{1}{n\alpha(n)r^{n-1}} \int_{B(0,r)} f(y) \, dy. \end{aligned}$$

Since $\rho(r)$ is continuous and differentiable, the Fundamental Theorem of Calculus gives:

$$\begin{aligned} u(0) &= \rho(r) - \int_0^r \rho'(t) \, dt \\ &= \int_{\partial B(0,r)} u \, dS + \frac{1}{n\alpha(n)t^{n-1}} \int_0^r \int_{B(0,t)} f(y) \, dy \\ &= \int_{\partial B(0,r)} u \, dS + \frac{1}{n\alpha(n)t^{n-1}} \int_0^r \int_0^t \int_{S^{n-1}} f(\omega s) s^{n-1} \, d\omega \, ds \end{aligned}$$

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